Unit 1

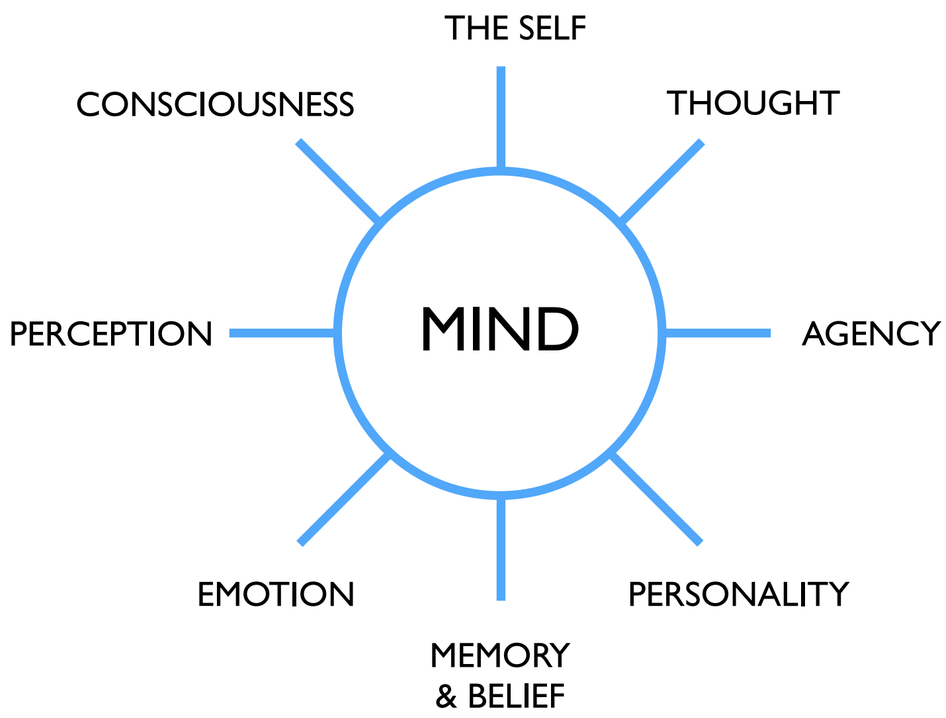
COMPUTATION

1. The Cognition-Body Problem

The Mind

What is a mind? What is *your* mind? There is no simple definition. The mind is the unity of **psychological** capacities, processes, and states.

Mental states are states of awareness, thinking, perceiving, feeling, hoping, imagining lying on the beach, and so on.



Dualism & Physicalism

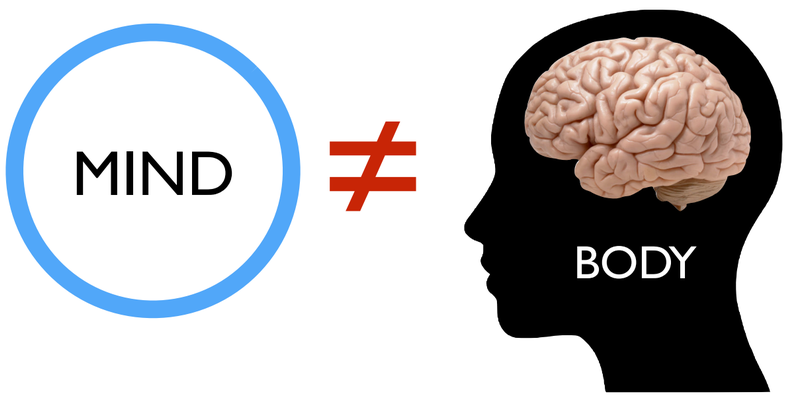
How do your own inner mental states fit into the objective physical universe? How do we reconcile mind and nature? The debate between two answers to this question is one of the deepest metaphysical debates in the history of philosophical thought.

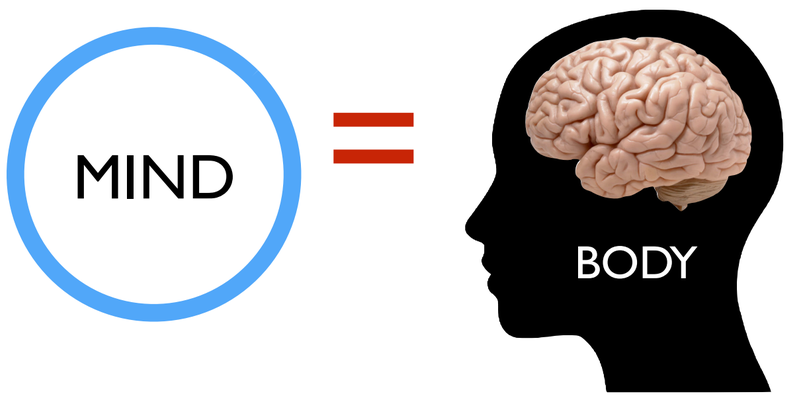
Dualism

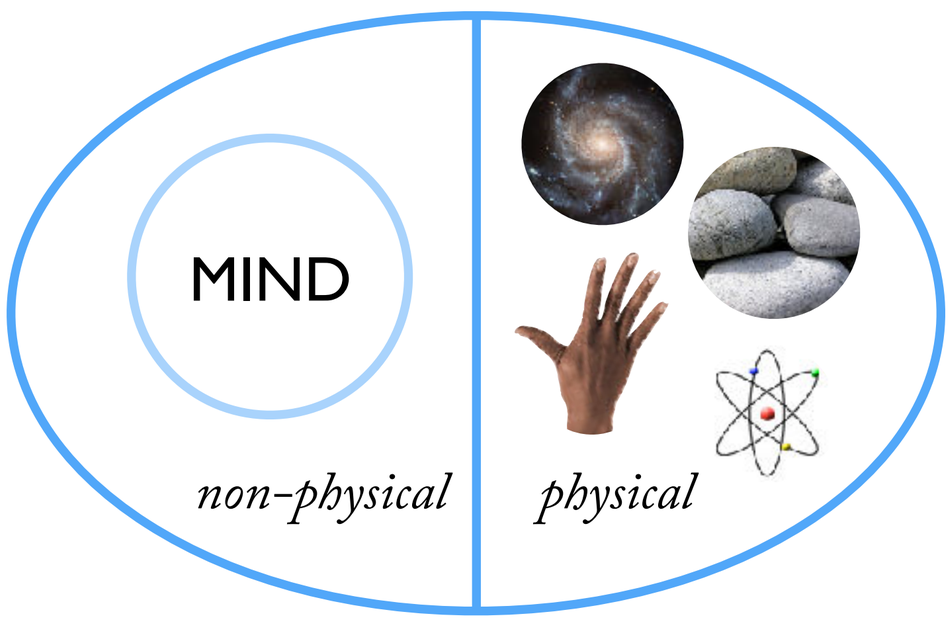
The mind is distinct from the body. No mental state is realized by a physical state.

Physicalism

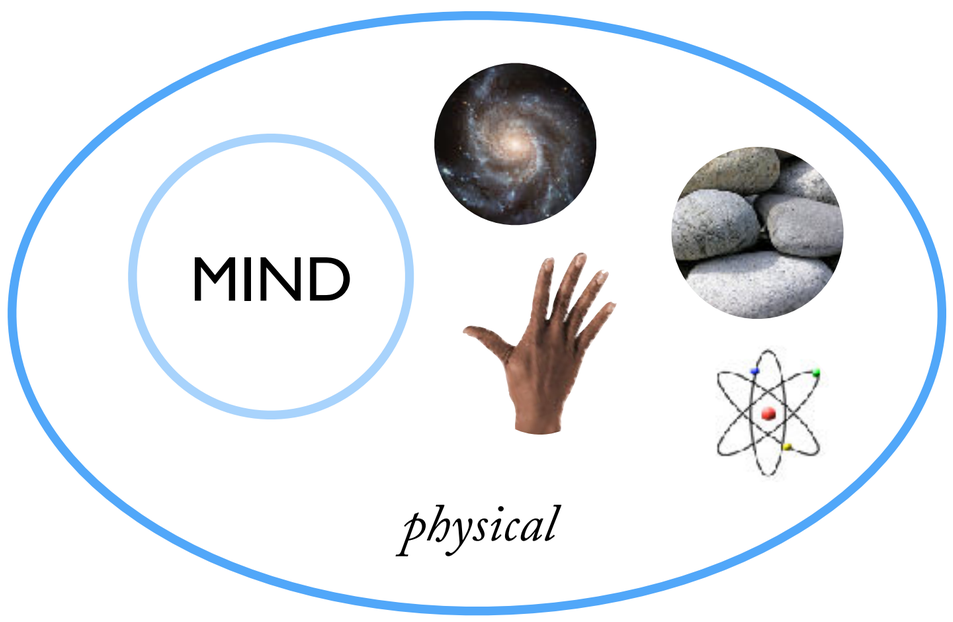
The mind is (part of) the body. Every mental state is realized by a physical state.







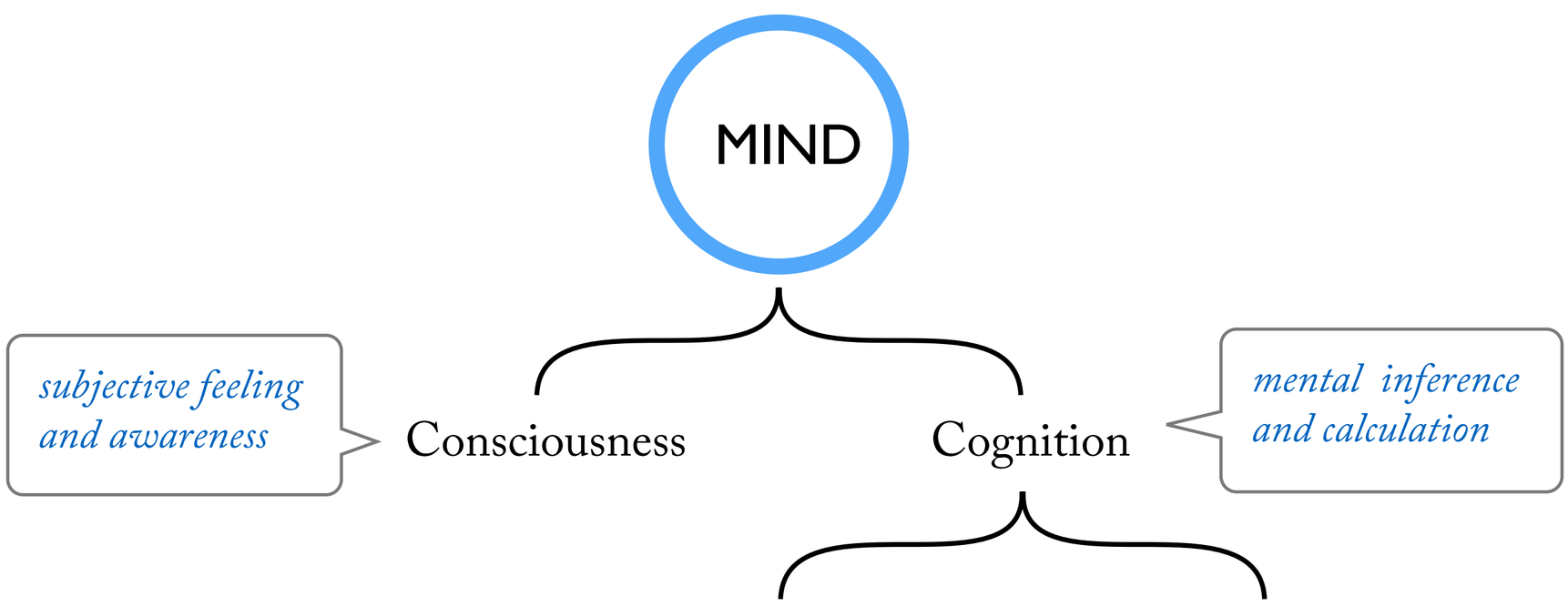
Reality according to **Dualism**



Reality according to **Physicalism**

Cognition & Consciousness

Two fundamental properties seem to distinguish minds from the rest of the non-mental universe. One is **consciousness**. The other is **cognition**.



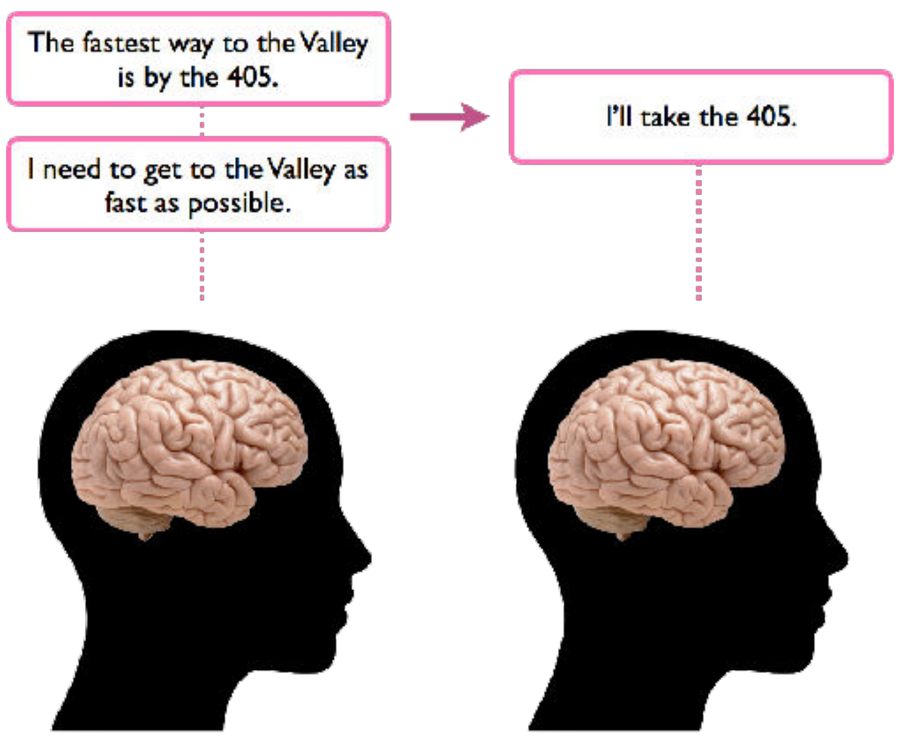
Contentful states: States possessing content.

content

Rational transitions: Rational transitions between contentful states.

content → content

A system has **cognition** if it undergoes rational transitions between contentful states.



A system undergoes **rational transitions** when it finds rational solutions to real problems.

The **content** of a state is whatever it is "about" or "of".

Individual words have particular individuals or properties as contents. Entire sentences have whole situations as their contents.

Examples of Mental States with Content

Believe: A believes *he is on Mars.*
Wish: B wishes *she could visit Mars.*
Know: C knows *she owns a G-Class.*
Think: D thinks *it's Tuesday.*
Learn: E learns *a dance move.*
See: F sees *a bicycle.*

Examples of Rationality

Logic: All X are Y. Q is X. → Q is Y.
Math: 3 + 4 → 3 + 4 = 7
Navigation: Maze → Path
Decision-Making: I want to see a movie. I like action movies. → I'll see an action movie.
Perception: [Light image] → It's a duck!
Prediction: Initial conditions → final conditions.

The Cognition-Body Problem

Leibniz (1646–1716) famously challenged the idea that cognition could emerge from purely physical elements. His objection is known as **The Windmill Argument:**

> One is obliged to admit that *perception* and what depends upon it is *inexplicable on mechanical principles*, that is, by figures and motions. In imagining that there is a machine whose construction would enable it to think, to sense, and to have perception, one could conceive it enlarged while retaining the same proportions, so that one could enter into it, just like into a windmill. Supposing this, one should, when visiting within it, find only parts pushing one another, and never anything by which to explain a perception. Thus it is in the simple substance, and not in the composite or in the machine, that one must look for perception.
>
> Leibniz 1714, *Monadology*

The argument has more or less the following form:

The Windmill Argument

Premise 1: Basic physical elements do not have cognition.

Premise 2: If a set of objects does not have cognition, then nothing composed of those objects alone has cognition.

Therefore: Nothing composed purely from basic physical elements has cognition.

Dualism is one conclusion to draw from this argument. If the premises are true, the argument implies that cognition is not physical. If it isn't physical, cognition must be the activity of a non-physical mind or soul.

Panpsychism is another possible reaction: deny Premise 1, and insist that even basic physical elements have cognition. In fact this was Leibniz' own view. (He called these basic minds "monads".) We won't discuss this interesting idea in this course.

Physicalism (or materialism) is the view that the mind (including cognition) *is* physical. Physicalists accept Premise 1, but deny Premise 2. They believe that non-cognitive parts *can* give rise to a complex whole which has genuine cognition.

The constructive challenge: physicalism is a promising *idea*, but how could you actually *make* a thinking thing from material parts? Where would you even begin?

The Computational Theory of Mind (CTM) is one version of physicalism. It seeks to understand the mind as the computational expression of the brain. CTM is more than just a philosophical idea; it is also the foundational assumption of cognitive science. The purpose of this course is to investigate CTM's answer to the cognition-body problem primarily as a philosophical and logical idea, and secondarily as an empirical hypothesis.

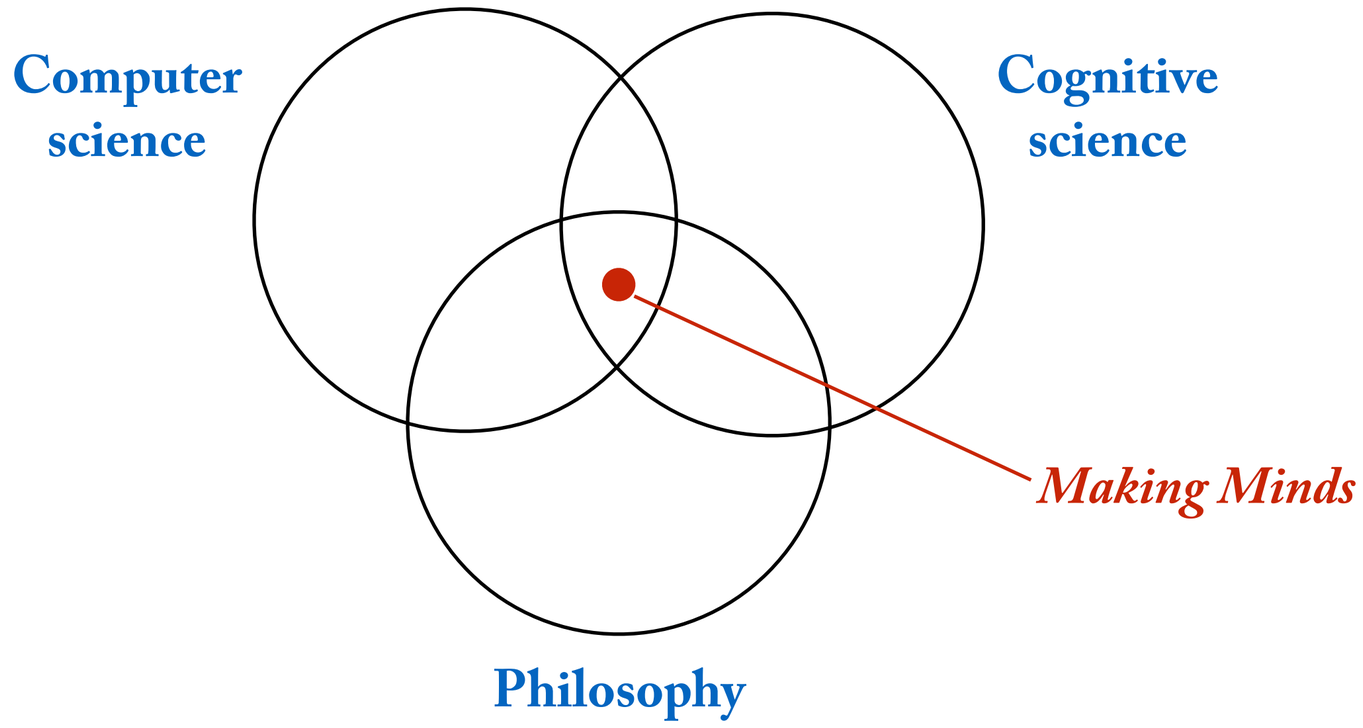
The Trajectory of This Course

This course is a sustained physicalist response to Leibniz' challenge.

Over 10 weeks we will try to see how far we can get, starting with the simplest non-thinking parts, and building towards more and more complex systems. The end goal is to design an organism whose physical construction is completely clear to us, but which seems to exhibit limited cognition. Our path to that destination is the computational approach to the mind.

The approach.

We will draw from computer science, cognitive science, and philosophy. We'll try to understand the big picture of the mind, by building detailed models of its key components. We'll use these simplified models to think about the complexity of the human brain.



Outline of the course:

Unit 1: Computation

We'll ask: what is computation? And we will construct simple logical computers from basic physical building blocks.

Unit 2: Levels

We'll expand our view of computers by considering higher levels of abstraction, and higher levels of computational power.

Unit 3: Cognition

We'll make the argument that computation in an embodied agent can give rise to genuine cognition.

Outline of Unit 1:

In this unit, we define the essential elements of computation.

- **Logic circuits:** our first machine.
- **Functions**: what computers do, defined precisely.
- **Representation:** how physical states relate to functions.
- **Computation:** a definition.
- **Computational theory of mind:** the conjecture at the heart of cognitive science.